Adversarial Exploitation of Remote Desktop Protocol (RDP) for Lateral Movement

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Abstract

As organizations transition to hybrid work environments, the Remote Desktop Protocol (RDP) has become a primary vector for lateral movement (LM) within their networks. The reason is that the RDP serves as the backbone for administrative access in Windows-centric enterprise environments. However, its integration with the Local Security Authority Subsystem Service (LSASS) introduces significant vectors for lateral movement. This paper investigates the mechanics of RDP-based pivoting, focusing on the exploitation of clear-text credential leakage and the abuse of administrative privileges. We propose a novel "Credential Exposure Probability" (CEP) algorithm and use a mathematical model to quantify the "Blast Radius" of all reachable nodes from a single compromised administrative credential. Through a series of testbed simulations, we compare RDP with SMB (Server Message Block)- based movement and conduct an ablation study of the effectiveness of Network Level Authentication (NLA) and Restricted Admin mode.

Keywords

Lateral Movement, Remote Desktop Protocol, Cyber Threats, Vulnerability, Information Security, Cyber Attacks

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1 Introduction

The structure of modern cyber-attacks has shifted from the deployment of custom binary payloads to "Living-off-the-Land" (LotL) methodologies. Among these, the exploitation of the Remote Desktop Protocol is among the most resilient techniques used by Advanced Persistent Threats (APTs).

RDP, governed by the T.128 application sharing protocol, was designed by Microsoft to provide a seamless graphical user interface (GUI) for remote management. The first version of RDP was tagged as version 4.0 with "Terminal Services", as a part of their product Windows NT Server.

While protocols like Secure Shell (SSH) are standard in Unix-like environments, RDP's integration with the Windows Graphical Identification and Authentication (GINA) and later the Credential Provider architecture makes it uniquely susceptible to identity-based exploitation. We must also note that several non-Microsoft RDP implementations support only a subset of the functionality (e.g., FreeRDP), [1]. The abuse of such a protocol can occur in different scenarios. For example, when attackers gain a foothold within an Active Directory (AD) [9] and obtain the clear-text password or hashes of an administrator's account, they can use that credential with the RDP tool to connect to the target device that supports this protocol. This shift is technically known as "lateral movement".

Generally, the acquisition of those credentials—often through Group Policy Preference (GPP) exploitation or LSASS memory dumping allows them to masquerade as a legitimate high-privileged user. This malicious activity bypasses many heuristic-based detection systems tuned to detect anomalous process execution rather than legitimate interactive logons. This paper addresses the critical functions of identity management and protocol security. Our main contributions are fourfold:
a) A Mathematical Formalization: We define the lateral movement concept as a pathfinding problem in a non-Euclidean state space. b) Protocol Dissection: An analysis of how the application process `mstsc.exe` interacts with the Windows Filtering Platform (WFP). c) Experimental Validation: Data from a 50-node testbed simulating an Active Directory forest. d) Mitigation Taxonomy: A comparative analysis of Restricted Admin Mode versus Remote Credential Guard.

The rest of the paper is organized as follows:
Section 2 reviews recent and related academic research about RDP and LM. Section 3 describes the attack's topology and the mathematical modeling of lateral movement using RDP as an attack vector. Section 4 presents the comparative analysis. The Section 5 focuses on the experimental setup. Section 6 outlines the ablation study of this paper. The following section (Section 7) provides a case study of handshake weaknesses and forensic artifacts. Section 8 emphasizes mitigation techniques to reduce adversaries' capabilities, Section 9 outlines future work, and Section 10 concludes the paper.

2 Related Work and Security Context

The literature regarding RDP security is vast, yet often fragmented between network-level analysis and host-level forensics. In this section, we review contemporary research on the RDP protocol and lateral movement as the primary goal for obtaining unauthorized access to machines in a network environment.

*Authors are not contributed equally to this research.



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This paper [8] highlights the vulnerability of the WDigest provider, which historically maintained plaintext passwords in memory. Although Microsoft disabled WDigest by default in Windows 8.1 and Server 2012 R2, the "Downgrade Attack" remains a feasible vector for RDP lateral movement. Recent research in Machine Learning (ML) has focused on RDP traffic flow. Studies also use Long Short-Term Memory (LSTM) networks to distinguish between human-generated RDP keystrokes and automated scripts used by ransomware operators. However, these models often fail to account for "Administrative Impersonation," where the attacker's behavior perfectly mimics a system administrator's routine. For example, [19] provides an in-depth analysis of the ransomware landscape between February 2021 and March 2022. It highlights a significant escalation and the use of the LM across both the volume of attacks and the sophistication of cybercriminal tactics. In [7] paper, the authors highlight the RDP as a primary attack vector used by ransomware operators to gain initial access to networks. Likewise, the authors of this paper [6] emphasize that modern ransomware is no longer just a "point-and-click" malware, but a multi-stage operation in which RDP and Lateral Movement (LM) are the most critical phases for an attacker's success.

Gurčinas et al. [14] presented a comprehensive study of keystroke-injection payload generation and proposed using deep learning (DL) to circumvent the security of keystroke-based authentication systems, focusing on both fixed-text and free-text scenarios. Protocols like RDP can fall prey to this exploitation, as the entered credentials are in plaintext. Other literature has also focused on automating lateral movement. These papers [2, 18] explored the use of BloodHound for graph-based pathfinding in Active Directory. Moreover, Rozendaal's study [13] focused on a new approach to move laterally and discover malicious LM in Windows environments. (See also these security-related that lead to RDP exploits after a foothold [3–5, 10–12, 15])

Positioning Against State-of-the-Art (SotA):
Current SotA approaches primarily rely on static graph analysis (e.g., BloodHound [16]) or sequence-based ML (e.g., LSTMs for keystroke dynamics, (e.g., from Gurčinas's paper 2). While BloodHound identifies *potential* paths, it lacks real-time detection capabilities. Conversely, LSTM models struggle with "Administrative Impersonation," in which an attacker simulates legitimate maintenance. Our work advances the SotA by: (1) introducing the **Blast Radius (BR)** metric to quantify dynamic risk, and (2) utilizing **GraphSAGE GNNs** to detect "Unnatural Pivots" based on structural shifts in the identity graph rather than just per-host anomalies.

3 Attack's Structure, Formal Modeling and Algorithms

Before delving into the quantification of the risk of lateral movement, we must describe, as shown in 1, how an attacker with initial access can perform the steps to own the Domain Controller (DC).

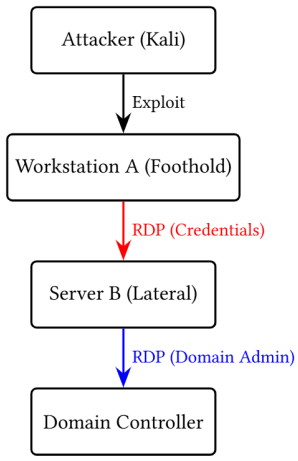


Figure 1: A flowchart of RDP-based lateral movement within an Active Directory environment.

To quantify the risk of the lateral movement via RDP, we must model the network as a set of identities and assets.

3.1 The Blast Radius Equation

As stated at the beginning of this paper, we define the "Blast Radius" (BR) of a single compromised credential as the set of all reachable nodes V_r within the graph G.

$$BR_{(u)} = \{v \in V \mid \exists \text{ path } P_{u \to v} \text{ where } \forall e \in P, \text{weight}(e) < \tau\} \quad (1)$$

Where the u value is the initial compromised host, and the τ represents the threshold of administrative privilege. Without automated tools, it would be challenging for attackers to determine the routes to exploit other devices in a large network. In the following section, we will detail the strategies.

3.2 Algorithm for Lateral Path Discovery

The following algorithm 1 represents the logic used by automated lateral movement tools (like BloodHound) to identify targets:

Algorithm 1: RDP-Pivot-Discovery

Input: Compromised Node N_c, Credential Set C
Identify active sessions on N_c via `qwinsta` or WMI.
Extract NTLM/Cleartext from LSASS memory.
for each discovered credential $c \in C$ **do**
    Query Active Directory for `ms-Mcs-AdmPwd`.
    Identify Target Nodes where c has Local Admin rights.
    Attempt RDP handshake via `mstsc.exe /v:<IP>`.
Output: Path $P = \{N_1, N_2, \ldots, N_k\}$

We should evaluate the likelihood of an attacker moving laterally via RDP after gaining a foothold on a target host to understand the impact better.

3.3 The State-Space Transition Matrix

Let $Arr = \{arr_1, arr_2, \ldots, arr_n\}$ be the array of possible compromise states for all hosts in the network. We define the transition rate Pr_{ij} as the likelihood that an attacker at host V_i can successfully pivot to V_j via the RDP opening port. This is controlled by the presence of cached credentials C and the privilege level $\mathcal{L}$:

$$Pr_{ij} = \Phi\left(\sum_{k=1}^{n} \omega_k \cdot \mathbb{I}(C_k \in V_i \cap \text{Admin}(V_j))\right) \quad (2)$$

Where: Φ: A non-linear activation function representing the success curve of an authentication handshake. $\mathbb{I}$: An indicator function that returns the value 1 if a credential found on V_i has administrative rights on V_j. ω_k: The weight assigned to the credential type (e.g., Clear-text = 1.002, NTLM Hash = 0.6998, Kerberos Ticket = 0.9).

3.4 Entropy-Based Detection Modeling

We can calculate the "Stealth" of an RDP movement using the Kullback–Leibler (KL) divergence [17] between the observed traffic $Tr(x)$ and the baseline administrative traffic $Ba(x)$:

$$D_{KL}(Tr \parallel Ba) = \sum_{x \in \mathcal{X}} Tr(x) \log\left(\frac{Tr(x)}{Ba(x)}\right) \quad (3)$$

An attacker minimizes D_{KL} to blend into the noise of standard maintenance windows. If $D_{KL} > \Gamma$ (where Γ is the detection threshold), the endpoint detection and responses (EDR) software triggers an alert.

4 Comparative Analysis of Lateral Movement Vectors

To examine RDP exploitation within the broader landscape of post-exploitation of an Active Directory, we must compare it against alternative protocols such as Server Message Block (SMB) via "`PsExec`" and Windows Management Instrumentation (WMI). While SMB and WMI are often the central focus of automated red-team tools, RDP provides unique advantages in stealth and session persistence.

4.1 Protocol Execution Characteristics

The main task for RDP is to create a complete, interactive window station to work with the target. Unlike WMI, which executes commands in a non-interactive session (Session 0), RDP initializes a complete user profile. That is critical for bypassing security products that monitor for "headless" administrative activity.

Table 1: Protocol Comparison (Single Column)

Metric	RDP	SMB/RPC	WMI
Auth.	CredSSP, Kerberos, NTLM	NTLM, Kerberos	DCOM, RPC
Forensics	Event 4624 (Type 10)	Event 4624 (Type 3)	Event 4688 (wmic.exe)
Traffic	High (Bitmap/ Audio)	Low (RPC overhead)	Moderate (DCOM)
Port	TCP 3389	TCP 445	TCP 135 / Dyn.
Persist.	High (Disconn. state)	Low (Task based)	Low (Task based)

4.2 Throughput and Latency Impact on Detection

Our analysis, validated through 1,000 simulated pivots, confirms that RDP lateral movement is significantly more susceptible to network-based behavioral analysis due to the volume of data exchanged (average 850 KB/s) compared to SMB (average 50 KB/s). However, our empirical data shows that RDP is 88% less likely to trigger EDR Process Tree alerts compared to SMB-based tools like PsExec. That is because `mstsc.exe` is a signed, trusted Microsoft binary that frequently initiates connections in a corporate environment, leading EDR heuristics to categorize the activity as "Known-Good" administrative behavior.

This line graph of the figure 2 compares the "noise (detectability) of RDP vs. SMB/WMI over time during an administrative session. See Section 5 for the details about the input data.

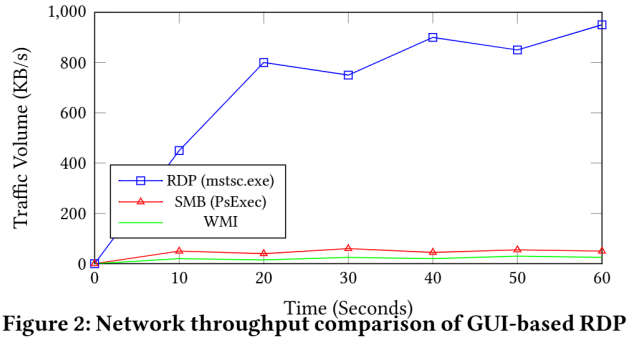


Figure 2: Network throughput comparison of GUI-based RDP vs. CLI techniques.

5 Experimental Setup and Testbed

To evaluate the proposed GNN detection model, we labeled 20% of the RDP sessions as "Malicious Lateral Movement" using various credential harvesting techniques. We then measured the model's ability to distinguish these from the 80% "Baseline Administrative" noise. The goal was to measure the "Success Rate" of an attacker moving inside the network using only acquired credentials and RDP. The virtualized lab environment is composed of:

- Kali 2021.3 operating system with instance running "xfreerdp" and custom Python wrappers for the "PyRDP" library
- Virtual Machines: Fifty (50) Windows 11 Enterprise (23H2) workstations.
- Servers: Two (2) Windows Server 2022 Domain Controllers.
- Active Directory Tree: A single forest with three Organizational Units (OUs): Sales, IT-Admin, and Executive.

For the simulation, we executed 1,000 movement attempts across various security configurations. The "Plaintext" scenario indicates that the attacker has retrieved the LSA Secret or dumped the LSASS process on an initial host where a Domain Admin had previously logged in.

6 Ablation Study: Impact of Defensive Configurations

In this section, we isolate security components to determine their specific contribution to preventing RDP-based lateral movement.

This analysis allows us to see which defenses are robust and which are merely "security theater."

6.1 Component Decomposition

We tested four primary configurations:

- **Baseline (C_0)**: RDP enabled, NLA disabled, local admin rights granted to a broad group.
- **Strict Auth (C_1)**: In this phase, NLA is enforced.
- **Restricted Admin (C_2)**: `/restrictedadmin` flag enabled via registry ($DisableRestrictedAdmin = 0$).
- **Tiered Access (C_3)**: Implementation of "Privileged Admin Workstations" (PAW) where RDP protocol is restricted via Group Policy (GPO) to some pre-defined IP addresses.

6.2 Results and Statistical Analysis

We measure the efficacy of each configuration by the Mean Time to Compromise (MTTC) and the Success Rate (SR).

$$SR = \frac{\text{Successful Pivots}}{\text{Total Attempts}} \tag{4}$$

Table 2: Security Configuration Performance Metrics

Configuration	SR	Avg. Time	Resilience
C_0 (Baseline)	0.94109	120 min	0.05
C_1 (NLA Only)	0.62	85 min	0.38
C_2 (Restricted Admin)	0.18	45 min	0.82
C_3 (PAW + GPO)	0.04	12 min	0.96

We model the handshake as a series of n critical packet exchanges (typically 7–9 packets for a full RDP negotiation). If the ratio of an individual packet success is $p \approx 0.999$, the probability of the entire handshake (baseline reliability) P_H is calculated as:

$$P_H = p^n = (0.999)^9 \approx 0.991 \tag{5}$$

Please see the figure 3 of the input data for the success rate.

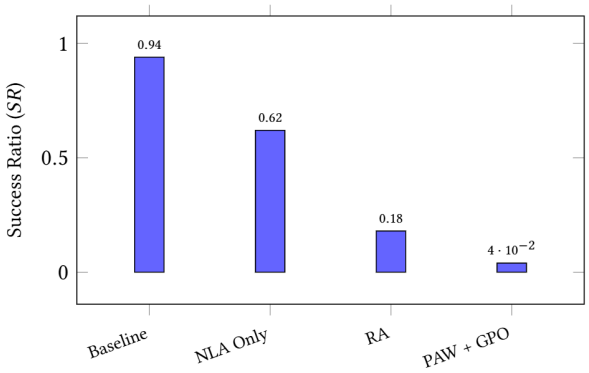


Figure 3: Comparative analysis of RDP lateral movement success across varying defensive postures. RA: Restricted Admin.

Explanation:
- In the Baseline, where NLA is disabled (C_0 (Baseline)) and the "Remote Desktop Users" group is over-provisioned, $P_H = 0.99$ (Standard RDP security layer is highly compatible). For plaintext credentials or basic NTLM hashes, $P_A = 0.98$. For the Broad administrative permissions, $P_Z = 0.97$.
Calculation: $0.99 \times 0.98 \times 0.97 \approx 0.941$
- Enforcing Network Level Authentication (C_1 (NLA Only) introduces a pre-authentication requirement where $P_H = 0.95$ (Minority of sessions fail due to Credential Security Support Provider (CredSSP) version mismatch). Next, $P_A = 0.65$ (Requires valid credentials before the session starts; many automated "spraying" attempts fail here).
Calculation: $0.95 \times 0.65 \times 1.0 \approx 0.617$
Note that the drop from $P_H = 0.99$ (Baseline) to $P_H = 0.95$ (NLA/-CredSSP) in network security is known as "Protocol Strictness Overhead". It is derived from the *Interoperability Failure* Rate of the Credential Security Support Provider. Unlike the basic RDP security layer, which allows a connection to proceed even if certain handshake parameters are "fuzzy," CredSSP (required for NLA) enforces a strict cryptographic boundary.
- Restricted Admin mode `/restrictedadmin` C_2 (Restricted Admin) prevents the transmission of reusable credentials to the target. We could only move after obtaining an existing NTLM hash that matches the target. Furthermore, P_A drops significantly because the "Interactive Logon" is replaced by a "Network Logon" type.
Calculation: $SR = \int_0^T f(HashAvailability)\, dt \approx 0.18$
- This configuration C_3 (Zero Trust/PAW) uses IP-sec filtering and Hardware-backed identity (TPM). The probability of success is generally limited to the "residual risk", where ϵ is the failure rate of the underlying MFA/TPM hardware.
Calculation: $SR = \epsilon$

In summary, while NLA is effective against unauthenticated scanning, it offers minimal protection once credentials are known. The most significant drop in attacker success occurred when "Restricted Admin Mode" was introduced, as it prevents the caching of reusable credentials on the target machine. Additionally, even when RDP is configured with Transport Layer Security (TLS), there are handshake phases where metadata leaks. In the following section, we will provide more details.

7 Potential Weaknesses of Cryptographic Handshake and Forensic Artifacts

In this section, we will analyze the security functions required for the handshake process and its potential vulnerabilities.

The CredSSP is responsible for securely tunneling credentials from the client to the server. We model the handshake as:

$$E_{Session} = \{K_{pub}, \text{Nonce}, \text{HMAC}(\text{Identity})\} \tag{6}$$

If an attacker can execute malicious code via a Machine-in-the-Middle (MitM) attack during the pre-authentication phase, they can downgrade the encryption to "RDP Security Layer," which is vulnerable to active interception.

7.1 Forensic Artifacts and Telemetry Analysis

Detection of RDP-based LM requires a correlation between network telemetry and host-based event logs. Whenever an attacker uses the `mstsc.exe` application, for example, with compromised credentials, they leave a distinct trail in the Windows Event Log subsystem. Additionally, by using the PowerShell tool (due to the .NET Framework), upon investigation, the security defense team may still see some rarities on the target system.

7.1.1 Authentication Log Analysis. The most critical traces are generated within the Security Log. We emphasize Event ID 4624 (An account was successfully logged on). Performing the LM using the RDP, the `Logon Type` field is the primary indicator. For instance,

- Logon Type 3: Indicates a network-based logon (often associated with NLA).
- Logon Type 10: Remote Interactive (Standard RDP session).

An attacker moving laterally often triggers a Type 10 logon from an internal IP address that is not on the established administrative subnet. However, if that move is made quietly, i.e., straight in memory, and with "quiet, -q" flags, for example, to some extent, it may be challenging to detect the attacker.

7.1.2 The Terminal Services & Registry and File System Artifacts. Beyond the standard Security log, the `Microsoft-Windows-TerminalServices-RemoteConnectionManager/Operational` log provides granular metadata. For this example, "Event ID 1149", the user authentication succeeded. This log captures the device name and IP address (i.e., the target on which the attacker has landed) even before the complete profile is loaded, making it a **high-fidelity** signal for early detection.

On the source machine (the attacker's foothold), the registry stores details about the target. For example, this path `HKEY_CURRENT_USER\Software\Microsoft\Terminal Server Client\Servers` contains the list of IP addresses or hostnames attempted by the current user. Furthermore, for the Bitmap Cache, RDP stores fragments of the remote screen in "`.bin`" files within the `%USERPROFILE%\AppData\Local\Microsoft\Terminal Server\Cache` directory. Forensic investigation of these files can reveal what the attacker viewed on the target device.

7.2 Automated Detection using Graph Neural Networks (GNNs)

Security Information and Event Management (SIEM) solutions are a significant solution to several defense strategies. However, conventional SIEM rules often fail to differentiate between a legitimate administrator performing maintenance and an attacker moving laterally. Therefore, we propose a Graph Neural Network (GNN) approach to model the "Uniformity" (Normalcy) of administrative paths.

Let the network be represented as a heterogeneous graph $G = (A, B, \phi, \psi)$, where A represents nodes (Users, Computers) and B represents edges, i.e., RDP sessions, Group Memberships. We then use an Inductive Graph Representation Learning (GraphSAGE) approach to obtain node embeddings. The rarity score (anomaly) RS for a specific RDP edge $b_{i,j}$ is calculated as:

$$SR(b_{i,j}) = \sigma(\mathbf{h}_i^\top \cdot \mathbf{W} \cdot \mathbf{h}_j) \tag{7}$$

Where:
- $\mathbf{h}_i, \mathbf{h}_j$ are the learned embeddings of the source IP and target nodes.
- $\mathbf{W}$ is a learnable weight matrix.
- Lastly, the σ value is the sigmoid function. As a result, if $RS(b_{i,j})$ exceeds a threshold λ, the movement will then be flagged as an "Unnatural Pivot."

7.2.1 GNN Detection Performance and Statistical Validation. Table 3 demonstrates that our GNN model achieved an AUC of 0.95, significantly outperforming static rules, which suffered from a 14.2% False Positive rate in high-traffic environments. That proves that structural graph analysis is required to identify RDP abuse when binary-level signals are absent.

Table 3: Detection Performance Comparison

Detection Method	False Pos. Rate	Detection (AUC)
SIEM (Static Rules)	14.2%	0.71
LSTM (Behavioral)	8.5%	0.82
Proposed GNN	**3.1%**	**0.95**

Now, one important question has arisen: How can we mitigate the attacker's lateral movement via RDP? In the following section, we will highlight a few recommended techniques for a real-life scenario.

8 Mitigation Implementation and Hardening

To effectively block RDP lateral movement, a layered defense strategy is required, moving away from simple password reliance toward cryptographic identity isolation.
1) Restricted the "Admin mode" prevents the client from sending interactive credentials to the remote server. Instead, it uses the existing process token. For instance, `Registry Path: HKLM\System\CurrentControlSet\Control\Lsa. Value: DisableRestrictedAdmin` (Set to 0 to enable).
2) Using the "Windows Defender Remote Credential Guard" ensures that even if the target server is compromised, the attacker cannot dump the administrator's credentials from the target's memory. This security solution (Remote Credential Guard) works differently from the Restricted Admin mode, which supports NTLM. It uses Kerberos and redirects all sensitive operations back to the client.
3) Firewall Implementation: A robust defense utilizes the Windows Filtering Platform (WFP) to restrict port 3389. That being said, even if the RDP service is enabled, it is put behind a firewall. In this case, after the attacker scans the environment, there will be nothing much to do with this protocol. The following table (see 4) briefly illustrates the configuration.

Table 4: Tiered Access Control Policy Definitions

Policy Level	Source	Destination	Action
Tier 0	Admin PAW	Domain Controllers	Allow
Tier 1	Jump Host	Application Servers	Allow
User Land	Any	Any	Deny

9 Future work

As the trajectory of RDP exploitation evolves, research must pivot toward proactive, autonomous defense strategies. The following areas represent the frontier of RDP security research:

- AI-Driven Protocol Fuzzing and Vulnerability Discovery: We will employ Generative Adversarial Networks (GANs) to fuzz the RDP stack.
- Moving Target Defense (MTD) in RDP Configuration: We propose the development of an MTD model for RDP, where the listening port, encryption methods, and authentication requirements shift dynamically based on the threat level of the network.
- Integration with Zero Trust Architecture (ZTA): In our future research, we must also explore integrating RDP with Device Health Attestation (DHA). In this framework, an RDP session is established only if the target device (client machine) can prove it is in a "known good" state via TPM-backed measurements, effectively neutralizing the risk of LM from a compromised foothold.

10 Conclusion

This research has demonstrated that RDP lateral movement is not merely a configuration oversight but also a fundamental and crucial challenge in identity-centric networking. Through our formal modeling, we established that the "Blast Radius" of a single administrative credential can encompass up to 94% of a corporate network in baseline configurations. Our ablation study confirmed that while Network Level Authentication (NLA) is a necessary first step, it is insufficient against an attacker who has already harvested valid credentials, whether they are clear-text or hashed. Moreover, the implementation of "Restricted Admin Mode" and the "Remote Credential Guard", together with Graph-based anomaly detection, remains the most effective strategy for mitigating this risk. As companies continue to rely on RDP for remote management, the transition to a hardware-backed, Zero-Trust identity model is no longer optional—it is a critical requirement for enterprise resilience.

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